PETW Events Bot Integrated Program Report
Events 事件分类与重建智能体集成报告

报告日期：2026-08-12 | 报告类型：bot-level local display/audit derivative | 证据截止：本地已核验 PETW Events、Step0 review、M0/M1 staging、BESSIE 观测包和 Job 3388245 display package

Bot-level work state
COMPLETE_FOR_REVIEWED_SLICE
五部分要求在本 Events 报告中闭合，报告不增加新的科学运行。

Scientific status
NOT_SCIENTIFICALLY_APPROVED
BESSIE 的实测阵风和对流外流过渡得到支持，下击暴流专属机制仍为 NOT_ESTABLISHED。

Promotion state
NOT_PROMOTED
无 EVT、Path C、验证风、Vhat_3s、design wind 或 design guidance。

结论先行。 Events lane 已形成可审计的候选记录、证据层、文献和 Wiki 资产、多源 BESSIE 观测重建、可探测性判断、CM1 接口包和失败关闭状态。当前证据支持 BESS 站在 2019-07-04 02:10 UTC 记录了 34.9 m/s 的实测阵风，并支持一次传播性对流外流或事件族尺度的传播性对流特征。当前证据不建立 BESSIE 的下击暴流专属机制，不建立事件真值，不建立 Path C，不建立验证过的 10 m 风或三秒阵风产品，也不产生设计量。

读者边界。 本报告把工作状态、证据类型、源文件、图形和表格放在同一份本地 HTML/PDF 中。已核验 只表示字节、哈希、结构或已声明方法范围内的检查通过，不表示科学机制已验证。部分核验 表示证据链仍有开放项。停止/阻断 表示执行或身份门关闭，不能被写成科学 null。

1. 报告范围、证据类别和状态词

科学问题是：已完成的 Events 工作能否被整理为一个自包含的 bot-level figure-and-table report，并在同一份制品中区分 bot 特征、Wiki 与 review 资产、工作状态、未来工作、实际观测结果、工程准备、停止或失败执行、display derivative、解释性结果和 PI scope decision。

本报告只读取本地 PETW vault 中已存在的状态文件、审计文件、review archive、Events 输出、BESSIE 观测包和 Job 3388245 display package。没有执行远程读取、HPC 或 scheduler 操作、CM1/WRF 新运行、Zotero 或 LitWiki registry 修改，也没有修改源报告、原始数据、review archive 或 predecessor artifact。复制到本报告目录的图形是 report-local display copies，源文件仍保留在原路径。

证据类别	本报告中的含义	可以支持	不能推出
已核验 文件/结构/哈希	文件存在，或声明的哈希、字节数、PDF、引用、图形、manifest read-back 在其范围内通过。	来源可追溯、制品可读取、报告可以复核。	科学机制验证、事件真值或 scientific approval。
来源声称 文献/Wiki	来自 source-preserving review、文献或 Wiki 的可归属表述。	方法背景、证据边界、候选输入。	把文献方法变成 PETW 已完成结果。
SCIENTIFIC_RESULT	直接观测或已完成模型输出在明确对象、时间和方法边界内的结果。	BESS WMAX、地面过渡、雷达扫描几何、Job 3388245 的 model-internal diagnostics。	跨对象等价、事件重建、验证风或设计量。
INTERPRETED_RESULT	基于多个已读证据的受限解释，保留竞争机制和开放项。	传播性对流特征在事件族尺度得到支持。	唯一局地机制、下击暴流真值、龙卷分类排除。
STOPPED / BLOCKED	执行、身份、资源、原始 QA 或授权门停止，输出可能为空。	保存失败原因、工程边界和下一决策点。	把未启动或未完成写成 scientific null。
PLANNED	来自审查或状态文件的未来工作，尚未自动启动。	形成新的 decision packet。	完成、验证、授权或科学提升。

必须保留的限制词。 相关源资产中的 NOT_ESTABLISHED、NOT_SUPPORTED、NOT_PROMOTED、PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION、[label] 和 not_scientifically_approved 不被本报告改写为更强状态。PARTIAL_ALIGNMENT / NOT_EXACT_REPLICATION 是 WRF lane 的跨包状态，在本 Events 报告中仅作为治理词汇保留，不作为 Events 结果使用。

2. Events bot characteristics and interfaces

Events bot 的职责是把异构的气象记录组织成可审计的候选证据和重建包。它不把一个 ASOS 或 Mesonet 记录直接称为下击暴流真值，也不把一个源生标签直接提升为 PETW mechanism label。它的实际工作单元是 source record、evidence record、candidate state、reconstruction packet 和 bounded handoff。

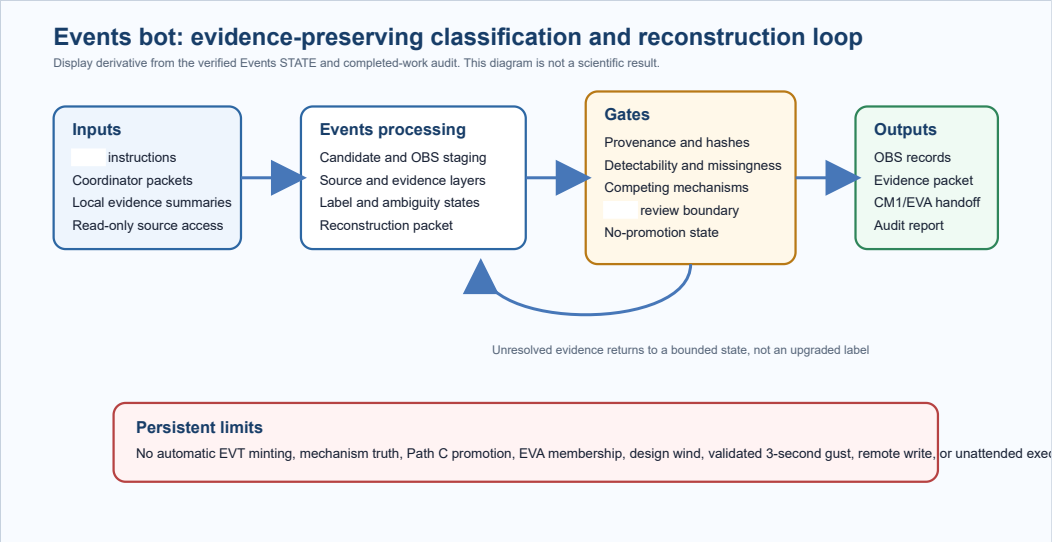


图 E0. Events bot 的输入、处理、证据门、输出和持续限制。该图是本报告的 display derivative，源语义来自 Events STATE.md 和共享完成审计，不包含新的科学结果。

能力	实际工作内容	当前证据	接口和边界
候选记录与标准化	保留源表 19 列、native integer knots、派生 mph/m/s、literal CST-6 to UTC 语义，并生成 provisional OBS 层。	M0/M1：879 source rows，879 OBS records、879 evidence records、879 narrative records；12 tests passed。	不生成 EVT，不把 879 rows 当作 879 independent events。
证据分层	把 source-native label、derived screening fields、independent evidence、mechanism hypothesis 和 promotion state 分开。	M1 readout、evidence-gap register、Events STATE。	`screen_score` 和 narrative-derived fields 只作 evidence-only screening features，带 leakage lineage。
多源重建	组合 Mesonet、Storm Events、NEXRAD Level II、MRMS、GLM、环境 profile 和邻站响应。	BESSIE 217-file acquisition manifest，26 compact entries，parent verification PASS。	缺失的观测几何、站点维护、局地雷达低层覆盖或独立证据必须保持开放。
可探测性与几何	计算雷达波束 AGL、站网距离和记录可用性，并将 detectability 与 mechanism state 分开。	BESSIE 三雷达 9 scans，velocity beam 1.568-3.031 km AGL；最近非 BESS 站约 28 km。	径向梯度是 20 km 局地拟合 proxy，不是完整水平散度；beam height 和单多普勒几何阻止分类排除。
竞争机制	并列保存传播性对流外流、下击暴流或微下击暴流、旋转污染、传播性多特征过程等假设。	BESSIE formal states 和 mechanism hypotheses。	“未建立”不等于“已排除”；“传播性特征支持”不等于唯一局地机制。
CM1 handoff	传递事件时间、位置、风速和方向证据、storm context、profile 可得性和缺口。	Events STATE 的 CM1 boundary；BESSIE provisional conditional package 和 Job 3388245 display。	CM1 决定 simulation worthiness；Events 不把 ASOS-only record 直接变成 CM1-valid input。
EVA/design-wind handoff	提供上游事件分类证据和不确定度，不计算 return period 或 annual-maximum tail。	Events STATE 的 EVA boundary。	不估计 return level，不生成 design wind 或 design guidance。
人类审批和入口	Lark direct bot active；LitWiki retrieval first-pass read-only；Coordinator packet route。	Events STATE：paired Lark DM、read-only retrieval route。	没有新 Telegram token；A3 execution 需要 对具体范围的明确批准，A4 默认 deferred。

2.1 当前自动化等级

Events 当前主要处于 A0-A2：A0 是规划和 schema，A1 是只读 ingestion 与 metadata，A2 是 supervised processing、QA 和 evidence packaging。已完成的 BESSIE 和 M0/M1 工作没有把 Events 变成闭环硬件控制系统。A3 只对明确批准的具体执行范围开放，A4 closed-loop 或硬件/远程自治保持 deferred。

Events -> CM1

事件重建 packet 包含时间窗、位置与站点几何、观测风速/方向、候选 storm morphology、radar/satellite/reanalysis availability、适用性理由和缺失信息。

接口状态：conditional input；不是自动 SIMCASE、CM1RUN 或机制真值。

Events -> EVA

Events 提供候选事件证据、源生标签和竞争机制状态。EVA 负责独立性、样本成员资格、尾部分析和设计量边界。

接口状态：upstream evidence only；不得由 Events 单独决定 return period 或 design quantity。

3. Wiki/RAG and review inventory

Events 的 scholarly assets 与 operational evidence assets 是两类不同对象。review archive 可以说明文献证据和方法边界，不能把 review build completion 当作 scientific approval。M0/M1 的 active retrieval 也保持 [label]

资产	实际内容和验证	状态	Claim ceiling
Events Step0 Review v2 中文	`ep0_Review_v2_20260726.pdf`	**ARCHIVED** 15 pages A4，55 source-bounded Events records plus one Moore 2024 interface source，56 eligible bibliography keys，2 vector schematics，deterministic Tectonic build。	`not_scientifically_approved`；文献证据和方法边界，不是 approved classifier、event truth、Path C 或 design quantity。
Events Step0 Review v2 English counterpart	`ep0_Review_v2_20260726_EN.pdf`	**ARCHIVED_CANDIDATE_BLOCKED** 21 pages，79 citation groups，56 unique keys，2 TikZ figures，0 tables，two deterministic builds and 21-page visual QA passed。	科学状态仍为 `not_scientifically_approved`。Stage 2/4 requested `gpt-5.6-sol` but observed parent runtime was `gpt-5.6-luna`，identity limitation disclosed。
Active Event Step0 Wiki/RAG	Active generation `g-20260731T225043Z-79730a429aea`，registry current SHA-256 `99cf784c05555ece8db0de6ab5f37a930ae99666171e0b179d4eeab59cfad66a`。Audit records 488 records，55 approved source-bounded Step0 evidence，433 not approved，source hash coverage 482/488。	**部分核验** Retrieval backend callable，but audit did not establish one compact governed-namespace inventory beside the archive。	只支持 source-bounded evidence design 和 evidence-gap identification；active retrieval result requires review。
M0/M1 legacy staging	`g/outputs/`	**COMPLETED** 879 provisional OBS，242 EPISODE_ID，140 multi-row episodes，maximum multiplicity 42；12 tests passed，deterministic rerun and Wiki entry/exit pin matched。	`event_id=null`，`truth_status=not_final`，event independence、mechanism、Path C、EVA、CM1 validity and confidence remain fail closed。
Existing event library assessment	Cook v2 ASOS gust events、Sui v2 storm objects、GridRad-Severe、NEXRAD Level II、Storm Events、IEM、MRMS、SWDI、historical field campaigns and synthetic/laboratory libraries。	**SOURCE AUDIT** Cook is candidate discovery and weak-label stratification；Sui is parent-storm context；no single source supplies full mechanism truth。	不得把 Cook `Thunderstorm`、Sui parent type、Storm Events event type 或 GridRad-Severe divergence 当作 PETW confirmed downburst truth。
BESSIE observational reconstruction	`E_Observational_Reconstruction_20260810_v1/`	**PASS BY SCOPE** 217 acquisition files，26 compact manifest entries；hash verification PASS；candidate pending review。	实测阵风、对流关联和 surface transition established；downburst-specific reconstruction `NOT_ESTABLISHED`；无 EVT。
Job 3388245 result display	`245_RESULT_DISPLAY_20260811_v2/`	**SOURCE-TRACEABLE / DISPLAY-ONLY** 241 NetCDF，5 spatial snapshots，60 s stats，7 source display figures，source/diagnostic manifest retained。	支持 model-internal ascent、precipitation、descent、local divergence and cold-pool diagnostics；Level-1 `NOT_SUPPORTED_UNDER_FROZEN_GATE`，不是 BESSIE event reconstruction。

3.1 文献与数据库的允许角色

资源	允许角色	禁止升级	关键限制
Cook ASOS v2	candidate discovery、source-native weak label、controls、QC/classification lineage。	同类地面特征分类器的 independent truth、无外部证据的 confirmed positive。	Development set 9,014 events；全国 Analysis set 超过 10 million events 为 TNNDT/XTNNDT machine labels；保留 `Checked`、`Class`、`XClass`、`TNNDT`、`XTNNDT`。
Sui v2	parent-storm context、radar-led candidate discovery、storm stratification。	local-mechanism truth、hard negative、validated wind。	GridRad v3.1 hourly，lowest grid 1 km，在该应用中主要为 reflectivity，覆盖四个 Texas cities。
GridRad-Severe	rapid radar candidate ranking。	10 m divergence、confirmed downburst label。	radar-derived gridded feature，起始高度约 0.5 km。
NEXRAD Level II	beam-height-aware radial velocity、polarimetric evidence、alternative-mechanism screening。	直接提供 event label。	观测源，不含机制标签；事件级解释必须先计算 beam geometry。
Storm Events / IEM / MRMS / SWDI	reports、surface recovery、storm evolution、contamination screening。	任何单源 mechanism truth。	event type、source quality、version epoch、station maintenance 和 product semantics 需逐项保留。

4. Work-state matrix: completed, stopped, blocked and planned

下表把工程准备、实时或失败执行、原始科学结果、display derivative、解释性结果和 PI scope decision 分开。一个 package ready 或 PDF build pass 不会自动改变 scientific status。

Work package	Evidence class	Current state	What is established	Claim ceiling / restriction
Events Step0 review v2	REVIEW_ASSET	Chinese archive complete; English counterpart archived candidate blocked。	文献 corpus、方法边界、source-bounded synthesis 和 review QA。	not_scientifically_approved；不是 classifier 或 event truth。
M0/M1 879-row staging	GENERATED_INPUT + EVIDENCE_STAGING	Completed，12/12 tests passed，two deterministic runs byte-identical。	879 source records round-trip；source episode multiplicity and missing metadata explicit。	不是 879 independent events；无 EVT、mechanism、confidence、EVA-ready 或 CM1-valid state。
M3 bounded anchor recovery	SOURCE_RECOVERY_EVIDENCE	35 files，873,948 recovered source bytes，cap 5,000,000 passed；JSON serialization repair verified。	bounded source-lineage、Mesonet station/site、MTS、compact radar/environment manifests。	OKEM、BESSIE、ERICK remain provisional OBS anchors；M4 not started。
BESSIE observation package	RAW/DERIVED_SCIENTIFIC_RESULT	Candidate pending review；event identity absent。	WMAX 34.9 m/s at 02:10 UTC；temperature decrease -5.2 C over 10 min；pressure rise 1.29 hPa；rain onset at 02:15；convective association and surface outflow transition established。	Downburst-specific reconstruction NOT_ESTABLISHED；propagating convective feature only supported at event-family level。
BESSIE radar geometry	VALIDATED_METHOD_COMPONENT within scope	9 scans from KFDR/KTLX/KVNX；raw along-beam gradient negative 9/9；dealias artifact count 1。	Velocity gates over BESS approximately 1.568-3.031 km AGL；no co-located low-level divergent core demonstrated by this screen。	Not full horizontal divergence；not categorical tornado exclusion；raw-priority rule must be retained。
Job 3388245 completed CM1 output	RAW_MODEL_RESULT + DISPLAY_DERIVATIVE	241 NetCDF，3600 s，5 spatial times and 60 s stats；display v2 source-traceable。	Model-internal ascent, precipitation, descent, local divergence and cold-pool diagnostics；peaks: wmax 22.4418 m/s at 2340 s, pratemax 0.00936125 kg m-2 s-1 at 2400 s, wmin -11.4658 m/s at 2520 s, divmax 0.00118535 s-1 at 2580 s, s10 max 2.98344 m/s。	VALID_DOWNBURST_CANDIDATE_LEVEL1: NOT_SUPPORTED_UNDER_FROZEN_GATE；not event reconstruction、validated downburst、validated 10 m wind、Vhat_3s or design load。
Job 3386168 BESSIE t=0	LIVE_EXECUTION / ENGINEERING_RESULT	Scheduler FAILED 20:0; CM1 model_rc 0；original QA FAIL；offline re-QA qualified；decimator processed 0。	96 original NetCDF retained for 0 and 60 s；model application normal termination and technical output evidence within stated scope。	Not event science；remote recovery not authorized and not ready；pilot blocked。
Fable 5 3600 s zc=1400 branch	ENGINEERING_PREPARATION / STOPPED	Stopped before package construction and scheduler submission due to trigger identity mismatch。	Actual TACC/Moore-author source sets zc=100 m while routed contract expected 1400 m。	No CM1 output、no technical QA、no scientific screening、no promotion。
Job 3387930 zc=100 branch	LIVE_EXECUTION / RUNTIME_GATE_STOP	Exactly one test-only and one real submission; scheduler FAILED 95:0; application not started。	Runtime identity gate false negative because symlink entrypoint and resolved target were compared as literal strings; other runtime predicates passed。	No history、stats、restart or model output；scientific screening NOT_EVALUABLE_NO_MODEL_OUTPUT；no retry authorized。
BESSIE control decision		Observed control retained; BESSIE rerun and package design stopped unless a new explicit decision reopens it。	Decision boundary and no-promotion constraints are durable。	Scope approval is not scientific approval and does not create a confirmed event label。

4.1 BESSIE observed surface sequence

UTC	TAIR (C)	PRES (hPa)	WSPD (m/s)	WMAX (m/s)	RAIN (mm)	Interpretation
02:00	26.9	951.28	14.4	19.8	0.0	Pre-transition observation.
02:05	25.0	951.65	15.9	26.4	0.0	Gust increasing before rainfall.
02:10	22.9	951.62	20.5	**34.9**	0.0	Measured peak; Storm Events 68 kt record is same BESS Mesonet source family.
02:15	19.8	952.94	14.5	18.6	3.56	Rain onset and post-gust thermodynamic transition.
02:20	18.7	952.38	9.1	13.6	7.62	Post-transition decay.

The BESS WMAX value is the maximum of 3-second observations during a 5-minute Mesonet period at 10 m measurement height. It is not a recovered continuous 3-second time series. Event-date station exposure and maintenance adjudication remain open.

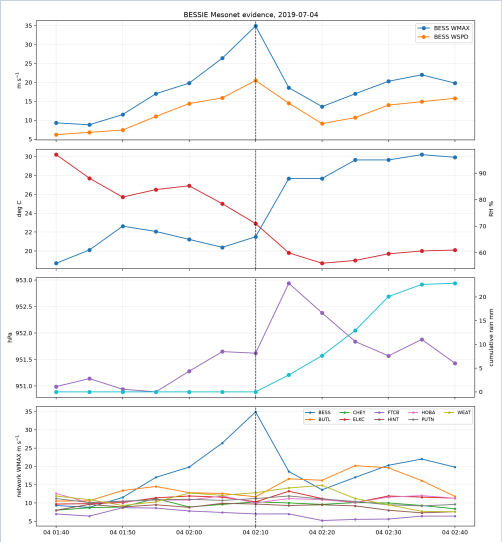


图 E1. BESS Mesonet transition. Source path: `E_MESONET_TRANSITION.png`. Evidence class: SCIENTIFIC_RESULT / SOURCE-TRACEABLE. Claim limit: supports the measured surface transition, not a unique local mechanism.

4.2 BESSIE storm-scale evidence and detectability

Evidence item	Observed value	Evidence status	Limit
MRMS core offset at 02:10	Maximum approximately 9.1 km east and 2.22 km north of BESS; centroid speed 31.32 km/h over the track window。	SUPPORTED	MRMS `bess_hsr_raw_value=-99.0` at 02:10 is missing/sentinel and is not used as low reflectivity。
Level II reflectivity	Three radars show strong convective core passage；50 km maxima about 62-64 dBZ in closest scans。	SUPPORTED	Reflectivity is storm context；not local downburst truth。
Raw radial gradient screen	9/9 raw local along-beam gradients negative；1 scan flagged as dealias artifact；dealiased values have nonzero sensitivity。	PARTIAL	20 km local derivative is a proxy；beam height 1.568-3.031 km AGL；not full horizontal divergence。
GLM lightning	40 flashes within 25 km and 44 within 50 km in the 02:10 five-minute bin。	SUPPORTED	Active convection only；low mechanism specificity。
Neighbor stations	Nearest non-BESS station about 28.27 km；responses delayed and spatially nonuniform。	LIMITED	Too sparse for a core-scale footprint or microburst/macroburst extent。
OUN profile	148.3 km away and 2.17 h before event；MUCAPE 2188.9 J/kg，MLCAPE 1860.7 J/kg，DCAPE 728.0 J/kg。	REGIONAL CONTEXT	Supports a regional moist pulse-severe environment；not event-local mechanism observation。

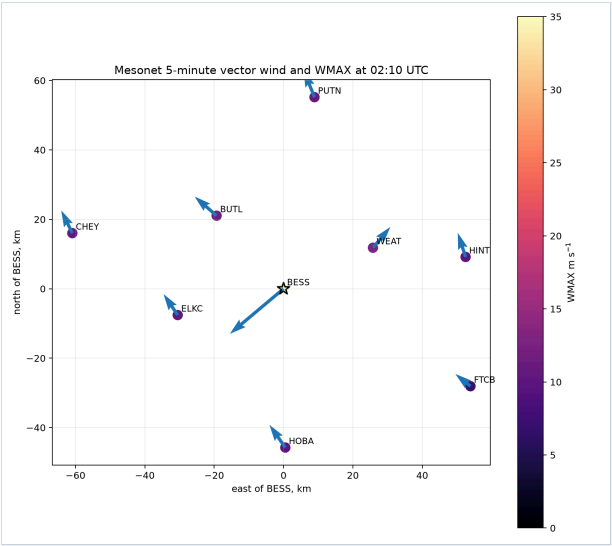


图 E2. BESS and neighboring station network at 02:10 UTC. Evidence class: SCIENTIFIC_RESULT / DISPLAY DERIVATIVE. Claim limit: network-scale response only.

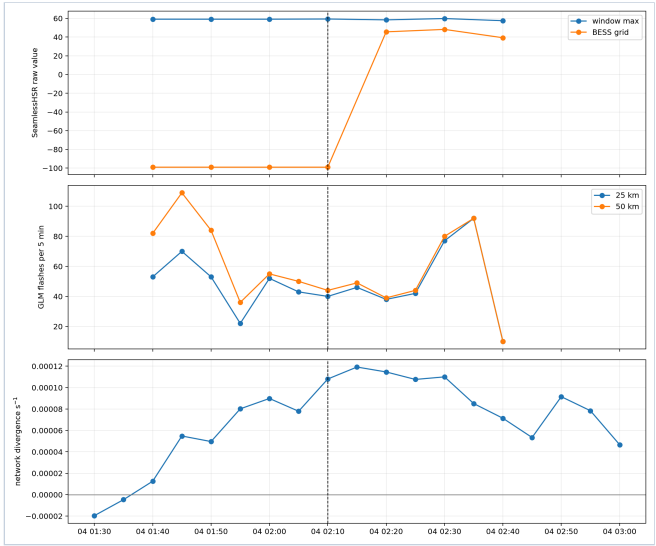


图 E3. MRMS and GLM storm-scale context. Evidence class: SCIENTIFIC_RESULT / DISPLAY DERIVATIVE. Claim limit: supports active convection and propagation context, not mechanism truth.

5. Integrated figure and table report

本节把现有 Events、BESSIE 观测和 Job 3388245 display assets 放在同一份视觉报告中。每个图形都在 report-local `figures/` 目录中有实际文件，并在 Figure Index 中记录源路径、语义、证据类别和 claim limit。图形不改变 source data，不做插值，不补造缺失时间。

5.1 BESSIE radar display derivatives

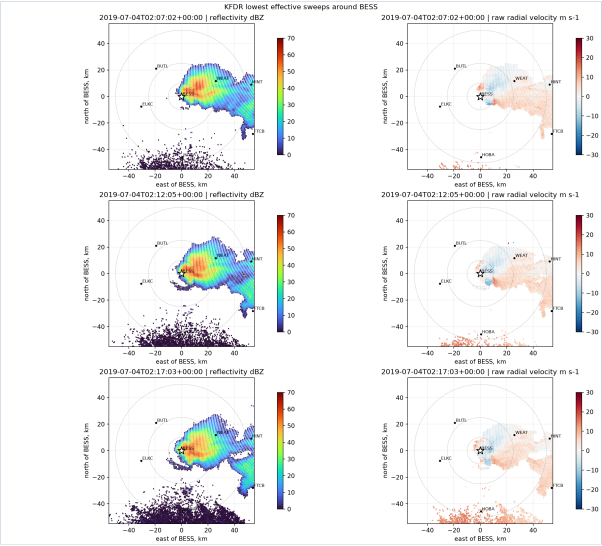


图 E4. KFDR Level II reflectivity/velocity display. Evidence class: DISPLAY_DERIVATIVE. Claim limit: beam-height-aware radar context；不构成近地面辐散证明。

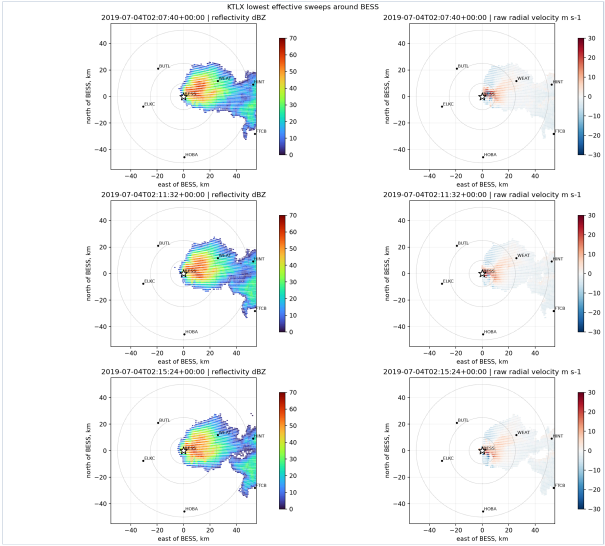


图 E5. KTLX Level II reflectivity/velocity display. Evidence class: DISPLAY_DERIVATIVE. Claim limit: storm structure and radial screening only。

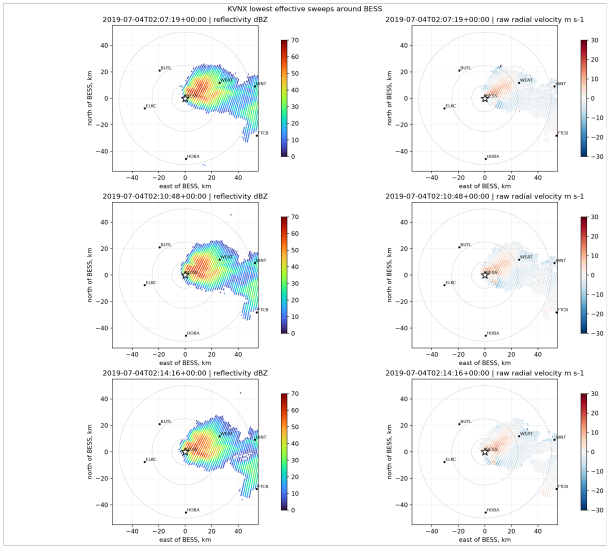


图 E6. KVNX Level II reflectivity/velocity display. Evidence class: DISPLAY_DERIVATIVE. Claim limit: three-radar context does not remove single-Doppler and beam-height limits.

5.2 Job 3388245 CM1 display derivatives

Metric	Value	Time	Meaning and limit
`wmax`	22.4418 m/s	2340 s	Model-internal maximum vertical velocity。
`pratemax`	0.00936125 kg m^{-2} s^{-1}	2400 s	Model-internal precipitation-rate maximum。
`wmin`	-11.4658 m/s	2520 s	Model-internal descent diagnostic；retained spatial cadence does not sample every peak morphology。
`divmax`	0.00118535 s^{-1}	2580 s	Model-internal 3-D divergence maximum；not observed surface divergence。
`s10` maximum	2.98344 m/s	60 s	Instantaneous near-ground diagnostic remains distinct from cumulative sws and observed WMAX。
Spatial output cadence	0, 900, 1800, 2700, 3600 s	Five snapshots	F2/F3 are discrete snapshots/positions, not a continuous trajectory。
Source output	241 NetCDF files	3600 s run	Source-traceable completed output and display derivative；not event reconstruction。

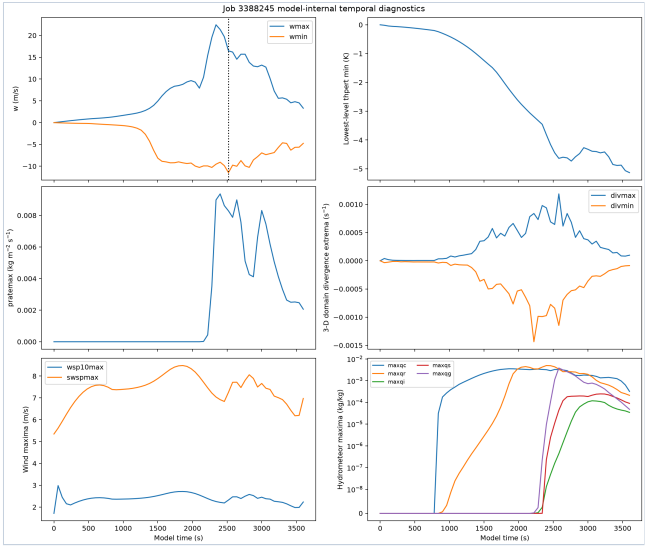


图 E7. Job 3388245 temporal diagnostics. Source path: `poral_diagnostics.png`. Evidence class: RAW_MODEL_RESULT / DISPLAY_DERIVATIVE. Claim limit: model-internal sequence only.

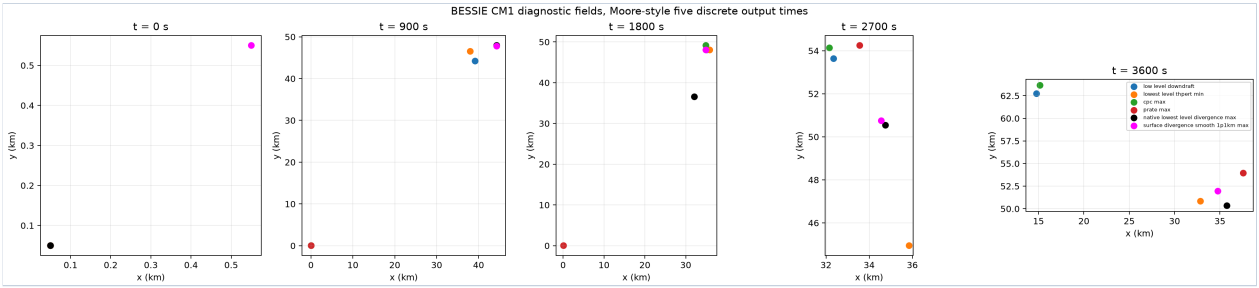


图 E8. Five discrete Moore-style diagnostic-center snapshots at 0, 900, 1800, 2700 and 3600 s. Evidence class: DISPLAY_DERIVATIVE. Claim limit: no continuous track and no interpolation.

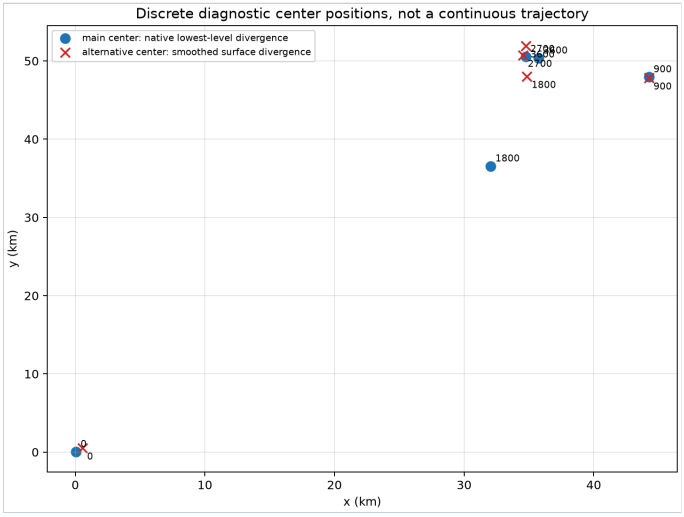


图 E9. Main and alternative feature-center positions. Evidence class: DISPLAY_DERIVATIVE. Claim limit: discrete positions, not a trajectory or unique propagating mechanism.

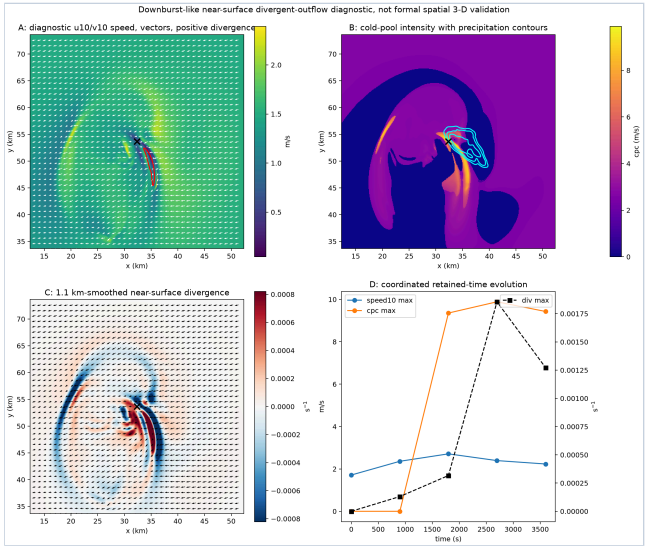


图 E10. 2700 s near-ground mechanism display. Evidence class: DISPLAY_DERIVATIVE. Claim limit: retained Level-1 gate remains `NOT_SUPPORTED_UNDER_FROZEN_GATE`.

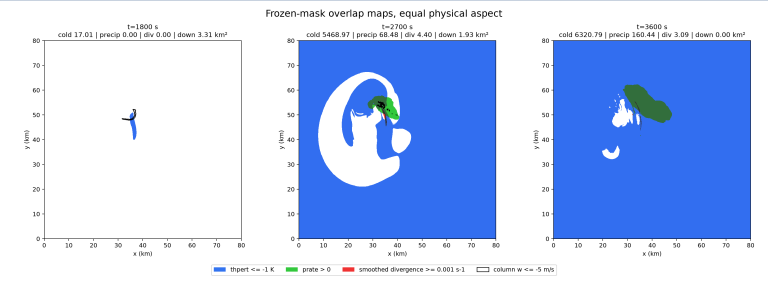


图 E11. Overlap masks at 1800, 2700 and 3600 s. Evidence class: DISPLAY_DERIVATIVE. Claim limit: mask semantics are source-defined; cumulative sws is not instantaneous 10 m wind and saturated masks are not independent corroboration.

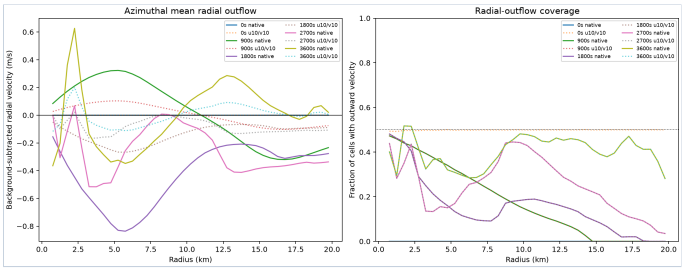


图 E12. Radial and azimuthal outflow diagnostics. Evidence class: DISPLAY_DERIVATIVE. Claim limit: model diagnostic coverage only。

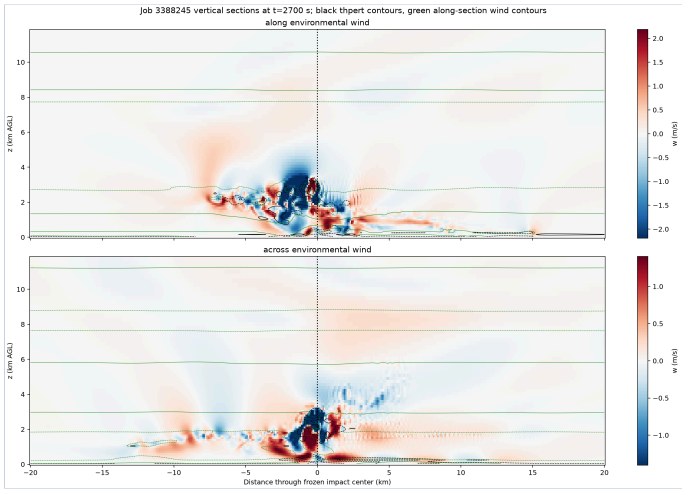


图 E13. 2700 s vertical sections. Evidence class: DISPLAY_DERIVATIVE. Claim limit: retained 2700 s minimum is near 2.10 km AGL； retained frames do not establish a surface-reaching descending core。

5.3 Figure index

ID	Report-local asset	Source path	Semantics	Evidence class	Claim limit
E0	`figures/F0_events_bot_loop.svg`	New report-local SVG derived from Events STATE and audit.	Bot input, evidence processing, gates and interfaces。	REPORT_DISPLAY	Architecture summary only; no new scientific result。
E1	`figures/BESSIE_MESONET_TRANSITION.png`	`.../SP1.../BESSIE_Observational_Reconstruction_20260810_v1/plots/BESSIE_MESONET_TRANSITION.png`	BESS surface sequence。	SCIENTIFIC_RESULT / DISPLAY	Measured transition; not unique mechanism。
E2	`figures/BESSIE_MESONET_NETWORK_0210UTC.png`	`.../plots/BESSIE_MESONET_NETWORK_0210UTC.png`	Station network geometry。	SCIENTIFIC_RESULT / DISPLAY	Sparse network; no core-scale footprint。
E3	`figures/BESSIE_MRMS_GLM_NETWORK_PROXY.png`	`.../plots/BESSIE_MRMS_GLM_NETWORK_PROXY.png`	MRMS/GLM storm context。	SCIENTIFIC_RESULT / DISPLAY	Active convection and propagation context only。

E4-E6	`figures/BESSIE_KFDR_LEVEL2_REFLECTIVITY_VELOCITY.png`
	`...KTLX...png`
	`...KVNX...png`
		`.../plots/`	Three Level II radar displays。	DISPLAY_DERIVATIVE	Beam-height-aware screening; not low-level divergence truth。

E7	`figures/F1_temporal_diagnostics.png`	`.../BESSIE_ZC100_JOB3388245_RESULT_DISPLAY_20260811_v2/figures/F1_temporal_diagnostics.png`	60 s model-internal temporal diagnostics。	RAW_MODEL_RESULT / DISPLAY	Conditional CM1 output only。
E8	`figures/F2_moore_style_five_times.png`	`.../figures/F2_moore_style_five_times.png`	Five discrete spatial snapshots。	DISPLAY_DERIVATIVE	Not a continuous trajectory。
E9	`figures/F3_feature_center_positions.png`	`.../figures/F3_feature_center_positions.png`	Discrete center positions。	DISPLAY_DERIVATIVE	Not a continuous track or mechanism separation。
E10	`figures/F4_2700s_figure5_style.png`	`.../figures/F4_2700s_figure5_style.png`	2700 s mechanism display。	DISPLAY_DERIVATIVE	Level-1 gate not supported。
E11	`figures/F5_overlap_masks_1800_2700_3600s.png`	`.../figures/F5_overlap_masks_1800_2700_3600s.png`	Mask overlap diagnostics。	DISPLAY_DERIVATIVE	Mask degeneracy and cumulative semantics retained。
E12	`figures/F6_radial_and_azimuthal_outflow.png`	`.../figures/F6_radial_and_azimuthal_outflow.png`	Radial/azimuthal diagnostics。	DISPLAY_DERIVATIVE	Diagnostic evidence only。
E13	`figures/F7_vertical_sections_2700s.png`	`.../figures/F7_vertical_sections_2700s.png`	2700 s vertical sections。	DISPLAY_DERIVATIVE	Does not establish surface-reaching core。

5.4 Table index

Table	Content	Source/evidence	Use
T1	Evidence categories and status words	Production packet, shared audit, PETW manuscript workflow	Interpretation control。
T2	Bot capabilities and interfaces	Events STATE.md and Master State	Bot characteristics and handoff boundary。
T3	Wiki/review inventory	Events review archive, English QA/readback, M0/M1, BESSIE package	Review/Wiki and display asset inventory。
T4	Work-state matrix	Completed audit and dated Events updates	Distinguishes results, preparation, stops, blocks and decisions。
T5	BESSIE observed sequence	BESSIE observational reconstruction JSON and nonradar analysis	Direct surface result with units and time。
T6	BESSIE storm-scale evidence	NEXRAD, MRMS, GLM and parent verification	Detectability and competing-mechanism limits。
T7	Job 3388245 display metrics	TEMPORAL_DIAGNOSTICS.json and result display package	Model-internal result and display semantics。
T8	Figure index	Report-local asset read-back and source paths	Figure provenance and claim ceilings。
T9	Future-work gates	Opus 5 Events progress review and Events STATE	Planned work only; no automatic start。
T10	Terminal status receipts	Report verification directory	Build, copy, hash, text and visual QA。

6. Future work and decision gates

以下内容是 planned work，不是本报告完成的工作。本报告在一个 report-production pass 后停止，不启动 revision、new scientific review、new simulation、remote action 或 successor task。

Stage	Planned purpose	Inputs and reused method	Required gate	Current state and stopping condition
S0	Freeze detectability、candidate ranking、positive evidence、hard-negative、ambiguous、event grouping and metric definitions。	BESSIE beam geometry、frozen gate pattern、alternative-mechanism structure。	review of the definition registry and any thresholds。	PLANNED。缺少可追溯阈值时留空，不猜测。
S1	Read-only detectability census and Cook v2 candidate extraction。	Cook source-native fields，station metadata，WSR-88D geometry，GridRad-Severe coverage，Storm Events，MRMS epoch。	authorization；source fields and machine/visual review state retained row by row。	PLANNED。站点元数据或 time mapping 无法确立时标记 not evaluable。
S2	Texas four-city overlap and stratified candidate ranking。	Cook + Sui + GridRad-Severe + Storm Events + Level II。	S0 registry frozen；Sui frontal/tropical strata retained；control classes retained。	PLANNED。时间、站点、雷达几何或 competing evidence unresolved 时保持 ambiguous。
S3	Depth reconstruction and truth-seed adjudication。	BESSIE acquisition/hash/readback、surface sequence、beam geometry、raw-priority radial screen、alternative hypotheses。	per-case decision；independent mechanism-sensitive evidence；no Cook/Sui-only adjudication。	PLANNED。证据不足的案例保持 ambiguous，不为凑数降低标准。
S4	Frozen blind evaluation of workflow performance。	S0 thresholds and S3 adjudicated set。	Minimum composition exists；event-level grouped split；ambiguous cases outside binary scoring。	NOT_STARTED。S3 composition不足时不报告 POD、FAR、CSI、precision、recall、F1 或 calibration。
BESSIE continuation	Remain observational convective-outflow control unless a new package-design phase is explicitly authorized。	Existing observation package and source-bound CM1 audit。	New decision；event-date siting/exposure；exact executable/source identity；explicit surface branch。	STOPPED。Do not mutate or resubmit Job 3388245。
M4 event evidence recovery	Additional anchor-specific source evidence after M3。	Recovered bounded source-lineage and open evidence-gap register。	Separate authorization。	NOT_STARTED。No event identifier or promotion before the gate。

> **最高优先级的未启动科学工作。** Events independent review identifies S1 detectability census as the next highest-value read-only action because it measures whether existing observations can support a mechanism-sensitive benchmark before expensive depth reconstruction. This is a planned routing statement, not authorization and not a new scientific result.

7. Claim matrix and final boundary

Claim subject	Current bounded wording	Evidence class	Not allowed
BESS measured gust	BESS recorded WMAX = 34.9 m/s at 02:10 UTC, defined as the maximum of 3-second observations during a 5-minute Mesonet period at 10 m。	SCIENTIFIC_RESULT	Continuous 3-second series or validated 3-second product。
BESS convective association	Surface transition, storm-scale radar context and GLM activity support convective association。	SCIENTIFIC_RESULT + INTERPRETED_RESULT	Single-source mechanism truth。
Propagation	A propagating convective feature is supported at event-family scale；gust front versus downburst-generated outflow versus multi-feature process is not distinguished。	INTERPRETED_RESULT	Unique local mechanism or categorical exclusion of rotational contamination。
Downburst-specific mechanism	`downburst-specific reconstruction: NOT_ESTABLISHED`。	SCIENTIFIC_RESULT / FAIL-CLOSED	Confirmed downburst、microburst、macroburst、VALID_DOWNBURST_CANDIDATE_LEVEL1。
Job 3388245	Completed traceable conditionally forced CM1 model output with model-internal diagnostics；display status `SOURCE-TRACEABLE/DISPLAY-ONLY`。	RAW_MODEL_RESULT + DISPLAY_DERIVATIVE	Moore 97-case reproduction、BESSIE event reconstruction、validated downburst、validated 10 m wind、`Vhat_3s`、design wind/load。
M0/M1 staging	879 provisional source observations and evidence records were created with explicit missingness and fail-closed states。	GENERATED_INPUT + EVIDENCE_STAGING	Completed event catalog、independent event count、approved labels。
claims	scope decisions constrain bounded work and stopping conditions。		Scope decision rewritten as scientific approval or publication permission。

Explicit no-promotion block. This report does not establish Events mechanism truth, validated downbursts, CM1 validation, Path C, `Vhat_3s`, design wind/load, CFD validation, exact WRF reproduction, publication permission or PI scientific claims. No EVT identity was minted. No remote write, solver rerun, scheduler action, Zotero change, LitWiki registry change or source-archive modification occurred for this report.

8. Source/evidence matrix and provenance

Source object	Exact path	Report use	Verification/status
Production packet	`fs/H-20260812-006_Events_CM1_WRF_Integrated_Report_Production_Packet.md`	Scope, five required parts, restrictions and stopping condition。	Read directly；source owner
Shared completed-work audit	`Updates/AU-20260812-006_Completed_Bot_Work_Audit_Against_Five_Part_Requirement.md`	Five-part status and gap closure。	Read directly；Events detailed figure/table report previously partial。
Events STATE	`jects/SP1_Event_Classification_Reconstruction/STATE.md`	Bot characteristics, interfaces, M0/M1, M3, BESSIE states and direct bot boundary。	Read directly；updated 2026-08-09。
Events Step0 Chinese review	`s/Reviews/Events_Step0_Review_v2_20260726/`	Review inventory, evidence levels, literature boundary。	15-page archived review; scientific status not approved。
Events Step0 English QA/readback	`s/Reviews/Events_Step0_Review_v2_20260726/verification/2026-08-11_Events_Step0_Review_v2_English_Final_QA.json`	English counterpart build, citation, forbidden-term, visual QA and identity limitation。	QA passes within declared scope; archive status blocked by exact model identity limitation。
M0/M1 completion manifest	`jects/SP1_Event_Classification_Reconstruction/10_Legacy_Staging/outputs/verification/COMPLETION_MANIFEST.json`	879-row staging counts, tests, deterministic rerun and no-promotion state。	completed；12 tests passed，remote_access_performed false，scientific_promotion_performed false。
BESSIE observation JSON	`E_Observational_Reconstruction_20260810_v1/BESSIE_OBSERVATIONAL_RECONSTRUCTION_20260810_v1.json`	Surface, radar, MRMS, GLM, environment, formal states and decision boundary。	Parent verification PASS；217 acquisition files and 26 compact entries。
BESSIE nonradar analysis	`E_Observational_Reconstruction_20260810_v1/derived/NONRADAR_OBSERVATION_ANALYSIS.json`	Surface sequence, neighbor timing and GLM counts。	Declared SHA-256 verified by independent review。
BESSIE NEXRAD analysis	`E_Observational_Reconstruction_20260810_v1/derived/NEXRAD_LEVEL2_ANALYSIS.json`	Beam height, raw/dealiased radial gradients and rotation-screen boundary。	Declared SHA-256 verified；9 raw gradients negative；one dealias artifact。
Job 3388245 result display	`245_RESULT_DISPLAY_20260811_v2/`	CM1 display figures, sidecar, temporal metrics, figure index, evidence matrix and package status。	Source-traceable/display-only；241 NetCDF；v2 repair preserves v1。
Events dated updates	`Updates/20260809_events_*.md`	t=0 terminal, Opus review, Fable corrections, trigger stop, zc100 runtime stop and progression state。	Read-only source summaries；each keeps engineering/scientific state separate。

9. Verification record and terminal status

本报告的验证记录位于同一 output root 的 `verification/` 目录。报告制品、复制资产、manifest、PDF text extraction、visual rendering and claim-safety scan are verified locally after generation。

Check	Method	Required result	Receipt
HTML validity and local references	UTF-8 readback，HTML asset path enumeration，all report-local image files and SVG checked。	**PASS**	`verification/report_verification.md`
Source-copy asset integrity	SHA-256 report-local copies compared with declared source paths for all copied images。	**PASS**	`verification/report_artifact_manifest.json`
True PDF build	Chrome headless print-to-PDF from the true HTML，not a placeholder file。	**PASS**	`verification/build_stdout.txt`
PDF text extraction	`pdftotext -layout` output read back；required sections and status strings searched。	**PASS**	`verification/report_text.txt`
Visual rendering	All PDF pages rendered to PNG；contact sheet and selected full-resolution pages inspected。	**PASS**	`verification/rendered_pages/ and verification/visual_qa.md`
Claim-safety scan	Forbidden terms, unsupported promotion strings, em dash, missing exact restrictions and absent required parts scanned。	**PASS**	`verification/claim_safety_scan.json`
Self-excluded manifest	Manifest lists report artifacts and excludes itself from its own hash set。	**PASS**	`verification/SELF_EXCLUDED_MANIFEST.json`
Source protection	Only report output root and one Agent_Update were written；no source report or raw package changed。	**PASS**	`verification/terminal_status.json`

> **Stopping condition reached.** The Events report package contains the five required parts, existing verified evidence, actual figures and tables, source paths, evidence classes, claim limits, QA receipts and terminal status. Work stops here. PI decides whether this report is accepted as the current internal Events bot-level program report. Scientific claim promotion, publication use, new remote runs and campaign expansion remain separate decisions.